

Benchmarking energy efficiency in vertical farming: Status and prospects

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ABSTRACT

Energy efficiency is a critical yet challenging key performance indicator for vertical farming. The absence of a representative benchmark has thus far hindered comparison with traditional agricultural techniques. This paper aims to establish the current status and future prospects of energy efficiency in vertical farming by exploring the interplay between plant physiology and facility design and operation. Energy use is primarily driven by the need to control the physical environment for crop productivity and quality, resulting in specific lighting and climate control requirements. While artificial lighting technologies are advancing towards the current biological limits of photosynthesis, climate control is increasingly influenced by plant physiological activity. An analysis of studies focusing on energy use in vertical farming reveals a current specific energy consumption of 10–18 kWh kg⁻¹ for lettuce production, corresponding to an energy use intensity of 850–1150 kWh m⁻² year⁻¹. Consideration of theoretical crop energy requirements, together with expected advances in equipment efficiency and operational control strategies, suggests a technical energy benchmark of 3.1–7.4 kWh kg⁻¹. Despite projected gains in energy efficiency, the current economic costs and environmental impacts associated with increased electricity use highlight significant challenges to the short-term energy sustainability of vertical farming.

1. Introduction

There is a growing demand for sustainable food production techniques that address the challenges of limited arable land and water scarcity, exacerbated by population growth and climate change. Two major research approaches are the modification of genetic potential for yield and the optimisation of crop yield in fertile environments [1]. The former involves engineering crop photosynthesis and yield potential [2] whereas the latter involves adapting indoor environments to plant requirements [3] and is termed controlled environment agriculture.

1.1. Controlled environment agriculture

Controlled environment agriculture is driving the transition from open field cultivation to greenhouse to vertical farm (Fig. 1).

Greenhouses are structures made of transparent materials in which crops are grown under sunlight and often controlled temperature and humidity [4]. They mimic the outdoor environment while protecting plants from the extreme conditions that could harm their growth. This extends the growing season and the range of products that can be produced in a given location compared to open field. Temperature and humidity control require different structures and equipment depending

on climate [5]. This results in typical climate-specific technological packages, tailored to address local environmental conditions and resource availability [6].

Compared to greenhouses, vertical farms, or plant factories, aim to optimise crop productivity through advancements in precision technology, data processing, and smart farming [7]. They consist of compact, multi-tier structures with opaque, thermally insulated, and airtight walls [8] that allow for higher yields per unit area. This comes at the cost of eliminating sunlight in favour of artificial lighting and results in the need for strict control of light, temperature, and humidity. Production facilities become highly reliant on advanced lighting and climate control systems. Such technology reduces the need for location-specific design due to the near independence from the external environment [9].

The global upscaling of vertical farming is a challenge for the future of agriculture, as vertical farms currently cover only a small portion of the market. With production consisting of approximately 90 % lettuce and 7 % strawberries [10], increasing the diversity of crops grown indoors is essential to sustaining fruit and vegetable production [11]. Success depends on crop features such as short harvest time and high harvest index [12], on economic aspects such as business models [13] or decision-making during planning and operation [14], and on resource use efficiency, encompassing water, nutrients, land, and energy [8].

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Nomenclature			
Abbreviations			
COP	Coefficient of Performance	h_{ev}	Heat of vaporisation of water, [kJ kg ⁻¹]
EUI	Energy Use Intensity	HI	Harvest index, [-]
HPS	High-Pressure Sodium	Q_{CC}	Climate control power requirement, [W]
HVAC	Heating Ventilation and Air Conditioning	Q_{env}	Heat exchange through envelope, [W]
LAI	Leaf Area Index	Q_{FIR}	Far infrared radiation, [W]
LED	Light Emitting Diode	Q_{light}	Heat generation of light source, [W]
PAR	Photosynthetic Active Radiation	Q_{plant}	Heat generation of plants, [W]
PPFD	Photosynthetic Photon Flux Density	Q_{vent}	Heat exchange through ventilation, [W]
SEC	Specific Energy Consumption	T	Temperature, [K]
Symbols		t	Time, [s]
c_v	Heat capacity at constant volume, [kJ kg ⁻¹ K ⁻¹]	V	Volume, [m ³]
DM	Dry matter content, [-]	X	Air water concentration, [kg kg ⁻¹]
E_{air}	Energy of air volume, [W]	X_c	Crop water concentration, [kg kg ⁻¹]
E_b	Biomass energy content, [kWh kg ⁻¹]	η_{can}	Efficiency of light supply to crop, [-]
E_L	Lighting energy requirement, [kWh kg ⁻¹]	η_{el}	Efficiency of electricity conversion, [-]
E_{PAR}	PAR energy requirement, [kWh kg ⁻¹]	η_{HVAC}	Efficiency of HVAC system, [-]
g_e	Evapotranspiration flowrate, [kg s ⁻¹]	η_{int}	Efficiency of internal processes, [-]
		η_L	Efficiency of lighting, [-]
		η_{PAR}	PAR share, [-]
		ρ	Air density, [kg m ⁻³]

1.2. Resource use efficiency

The transition from open field to controlled environment agriculture leads to significant changes in resource use efficiency. Fig. 2 shows the resource use efficiency for lettuce production in open field, greenhouse, and vertical farm in terms of grams of fresh weight per litre of water consumed, daily productivity per unit of land, and grams of fresh weight per unit of energy used. While water and land use efficiency improve with controlled environment agriculture, these gains come at the expense of energy consumption [15], due to increased lighting and cooling demands.

In the available literature, it is reported that energy use efficiency in vertical farming decreases on average by twenty times compared to open field cultivation [16]. This has important consequences on the environmental impact of crops, as energy accounts for about 90 % of the carbon footprint [17]. It is indeed estimated that the carbon footprint can increase from values of 0.2 kgCO₂ kg⁻¹ in open field to 20 kgCO₂ kg⁻¹ in vertical farms [16].

1.3. Motivation and novelty of the work

Energy efficiency in controlled environment agriculture is receiving increasing attention [18], but the lack of standardised performance metrics remains a fundamental barrier to its widespread adoption [19]. Although various studies have attempted comparisons [20–22], no representative benchmark has emerged. Establishing a performance standard is key to determining the overall sustainability of vertical farming compared to traditional agricultural techniques. It requires consideration of trade-offs between environmental impacts [23] and economic costs [24] and should include expected advances in equipment efficiency [25] and optimal control strategies [26], providing a clear performance target to be pursued in real operating facilities.

A key novelty of this study lies in bridging elements of plant physiology and facility design and operation, tracing crop energy requirements back to the specific energy demands of the equipment for lighting and climate control. This integrative approach provides a comprehensive understanding of energy use in vertical farming. In order to standardise energy use, this work organises the relevant information on vertical farming available from literature studies and existing facilities, and it derives a technical energy benchmark that reflects technological advances in equipment efficiency and operational control

strategies. The method is applied to lettuce production in vertical farming to allow for comparison with previous benchmarks, but generalisation to other crops is straightforward.

The paper is structured as follows. Section 2 discusses the control of the physical environment as the primary driver of energy use. Section 3 considers lighting and the progress of artificial lighting technologies, while Section 4 focuses on climate control and the increased influence of plant physiology. Section 5 establishes current and anticipated energy benchmarks based on literature data and simple yet robust energy considerations. Lastly, some concluding remarks on the potential uptake of vertical farming are made based on the relevant findings.

2. Control of the physical environment

In vertical farming, environmental control is performed to ensure that the plant has sufficient resources such as CO₂, water and light for the photosynthesis process, and that environmental conditions such as temperature and humidity are optimised for the biological processes involved in plant growth. It mainly involves the control of light, temperature, humidity and CO₂ [27], the former being the most complex, as it must be controlled in terms of quality, intensity, and duration [28].

Light quality refers to the spectral distribution of light and is measured in terms of Photosynthetic Active Radiation (PAR), the range of wavelengths in which photons can be used for photosynthesis. The PAR range is traditionally defined as the interval of wavelengths between 400 and 700 nm [29], though it could be extended to 400–750 nm to include far-red photons [30]. Light intensity is measured in terms of Photosynthetic Photon Flux Density (PPFD) or Daily Light Integral (DLI), depending on the time scale considered, while light duration is measured in terms of photoperiod.

Cardinal temperatures are used to summarise the influence of temperature on different growth events for most plants [31]. They define the minimum and maximum temperatures for plant growth and the single or range of optimal temperatures at which growth rate is maximal, and they differ for light and dark periods. For humidity, relative humidity is more important than absolute humidity because it regulates transpiration rates and affects the water balance of the plant [32]. The vapor pressure deficit is often used, as it is the driving force of water loss from a leaf [33].

Controlled environment agriculture aims to optimise crop productivity and quality by identifying ideal combinations of controlled

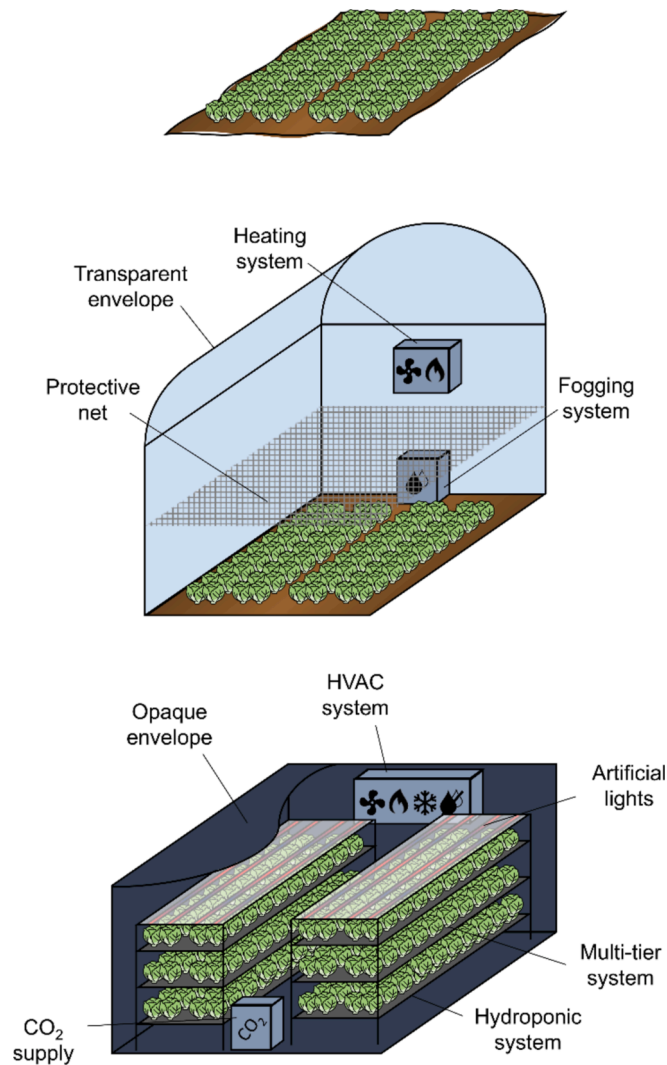


Fig. 1. Cultivation in open field (top), greenhouse (middle), and vertical farm (bottom).

parameters [35]. Fig. 3 shows the optimum PPFD, photoperiod, temperature, and relative humidity for lettuce and strawberry. The optimal range of PPFD is narrower for lettuce because of the large number of studies on this crop. The optimal temperatures for the two crops are the same at night but different during the day. Furthermore, Fig. 3 emphasises the importance of transition periods between day and night temperatures, often overlooked yet critical for the operation of climate control systems [36]. Variation in optimal values are important also for the same crop at various developmental stage.

Identifying optimal recipes of controlled parameters is a complex task due to the many influencing factors. The influence on crop growth has been characterised either as a combination of the interactions of complex physical phenomena [37] or simply modelled as separate contributions. In the latter case, the overall effect can be obtained either by considering the value of the limiting factor, as done in Ref. [38], or as the product of each specific contribution, as done in Ref. [20]. The resulting influence of each parameter is greater when it plays the role of limiting factor.

Such performance maps similar are currently spreading in the academic literature, both for total crop yield [24] and for individual biological processes [7], but they have several shortcomings due to the complex actual plant response to abiotic stress. They are often based on averaged growth parameters, which should instead be considered as time- and parameter-based functions, and often lack of validation on a

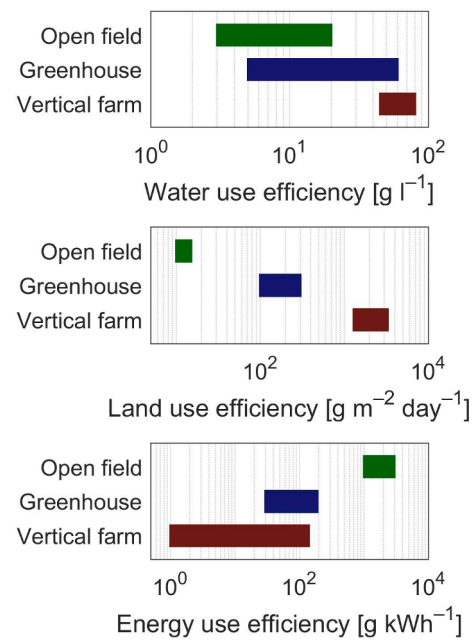


Fig. 2. Resource use efficiency for lettuce production: water (top), land (middle) and energy (bottom). Data from [16].

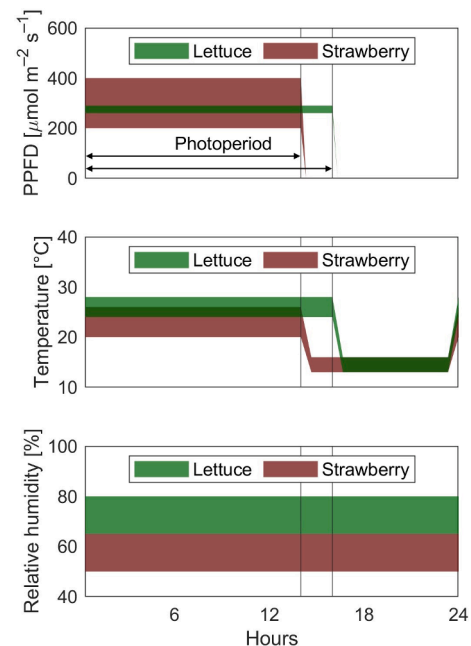


Fig. 3. Optimal values of PPFD (top), temperature (middle) and relative humidity (bottom) for lettuce and strawberry. Data from [34].

sufficiently wide range of environmental conditions [39].

The control of the environmental parameters leads to significant energy demand for lighting and climate control. The quantification of the energy consumption requires consideration of the complex interplay between crop energy requirements and facility energy use, including features of the conditioned space and of the conversion devices [40].

3. Energy use for lighting

Lighting is required for the conversion of energy into plant biomass. The crop energy requirement for lighting can be evaluated by considering the amount of chemical energy contained in the crop and the ef-

efficiency of fixation. As described as in Equation (1), the PAR requirement E_{PAR} in kWh kg^{-1} can be expressed as a function of the energy content of plant biomass E_b in kWh kg^{-1} , the efficiency of the internal processes for PAR conversion η_{int} , the dry matter content DM and the harvest index HI .

$$E_{PAR} = \frac{E_b \cdot DM}{\eta_{int} \cdot HI} \quad (1)$$

A rule of thumb for the energy content of dry biomass is $20 \text{ MJ kg}^{-1}_{\text{dry matter}}$ [8], but detailed assessments tailored to specific crops can be based on the composition and associated heat of combustion [41]. The efficiency of PAR conversion depends on several processes, as shown in Fig. 4 for sunlight PAR. At best, only about 10 % of PAR is converted into phytoenergy due to the low efficiency of photosynthesis. The values considered are only slightly lower than those calculated in Ref. [42]. The dry matter content and the harvest index are the factors that vary more significantly on the crop.

Lighting energy sources can be more or less tailored to crop requirements. Fig. 5 compares the relative action spectrum of sunlight, a

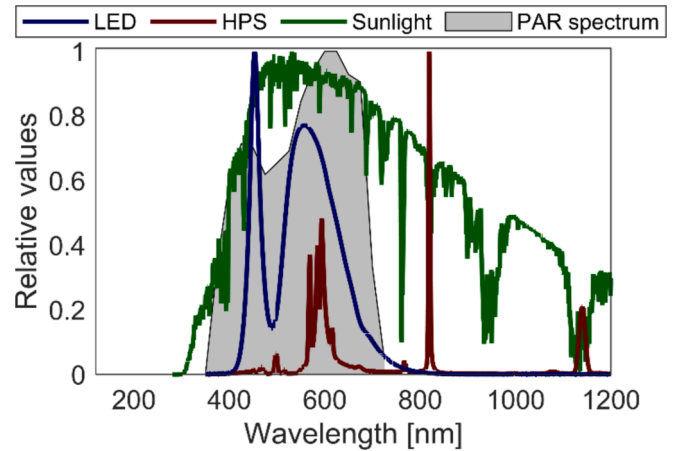


Fig. 5. Relative action spectrum of sunlight, HPS lamp, and LEDs, and representative quantum yield of crops. Data from [43–45].

High-Pressure Sodium (HPS) lamp, a white Light Emitting Diode (LED) and the typical PAR spectrum of crops [29]. The selectivity of LEDs allows them to match crop requirements better than both sunlight and HPS lamps.

For artificial lighting, the losses to be considered are the radiation other than PAR, typically near infrared radiation, the energy not successfully delivered to the plants, and the efficiency of the electricity conversion. This is summarised in Equation (2), where the lighting energy consumption E_L in kWh kg^{-1} is obtained from the PAR energy requirement E_{PAR} in kWh kg^{-1} , and the lighting efficiency η_L , which can be expressed as the product of the electricity conversion efficiency η_{el} , the efficiency of light delivery to the canopy η_{can} , and the proportion of PAR to the total light supply η_{PAR} .

$$E_L = \frac{E_{PAR}}{\eta_L} = \frac{E_{PAR}}{\eta_{el} \cdot \eta_{can} \cdot \eta_{PAR}} \quad (2)$$

The placement and distribution of the light sources to match the morphology of the crop is very important. HPS lamps provide a wide, uniform output suitable for overhead lighting in large spaces, while LEDs deliver a more focused pattern suitable for both overhead lighting in small spaces and intra-canopy lighting [46], as they can be placed closer to crop tissues. With proper configuration, approximately 90 % of the energy can be successfully delivered to plants by both HPS and LED installations [47]. The electricity conversion is around 60 % for both lamps, but HPS lamps only emit around one third in the PAR range, whereas LEDs emit 96 % in the PAR range [48].

Fig. 6 shows the conversion of electricity to PAR for HPS and LED. The overall efficiencies are approximately 33 % and 50 %, respectively. These are only slightly higher than the current upper limits for commercial horticultural lamps [49], although efficiencies of up to 90 % are expected for LEDs [50]. The efficiency of sunlight is around 48 % with losses only due to non-PAR radiation. As a result, the greater efficiency of LEDs is one of the advantages of artificial lighting over natural lighting. As the LED spectrum matches the crop quantum yield better than sunlight, as shown in Fig. 5, additional improvements in the conversion efficiency could be obtained.

The overall lighting efficiency is also influenced by the developmental stage of the crop. Inefficient lighting supply can be due to the difference between PPFD for maximum productivity and maximum efficiency and to the low light use efficiency during earlier growth stages when the Leaf Area Index (LAI) is low [51]. This can be solved with passive or active techniques, such as adapting plant spacing and using reflectors [52], or tailoring lighting intensity based on the crop actual requirements [26,53].

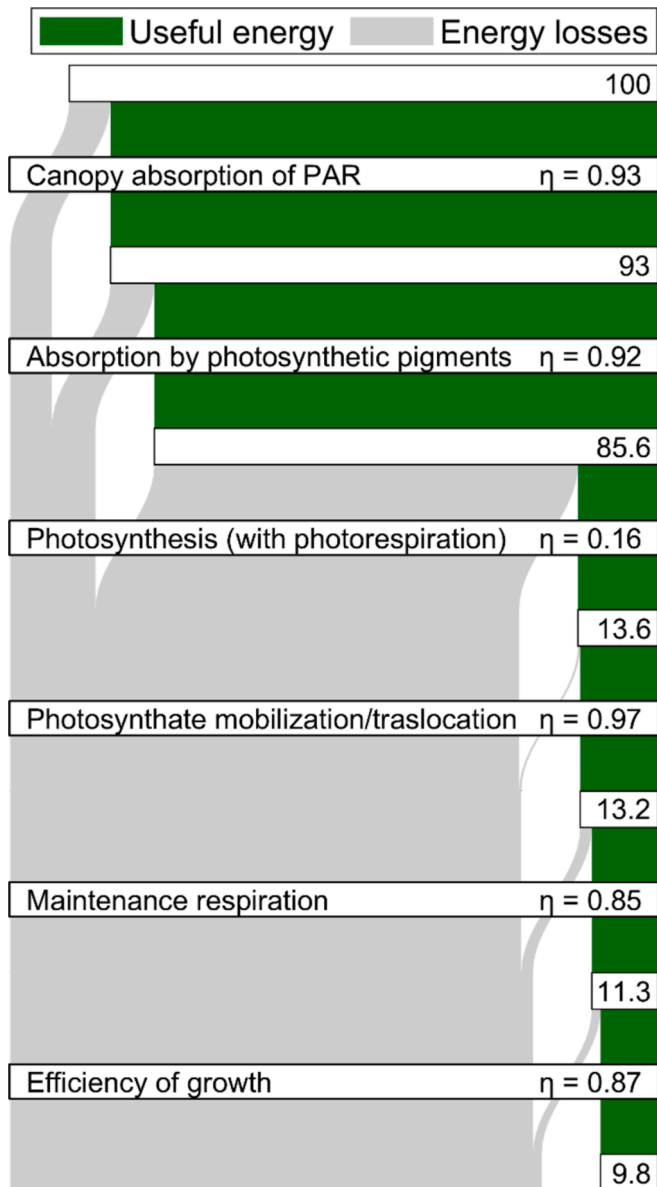


Fig. 4. Maximum conversion of sunlight PAR into phytoenergy. Data from [41].

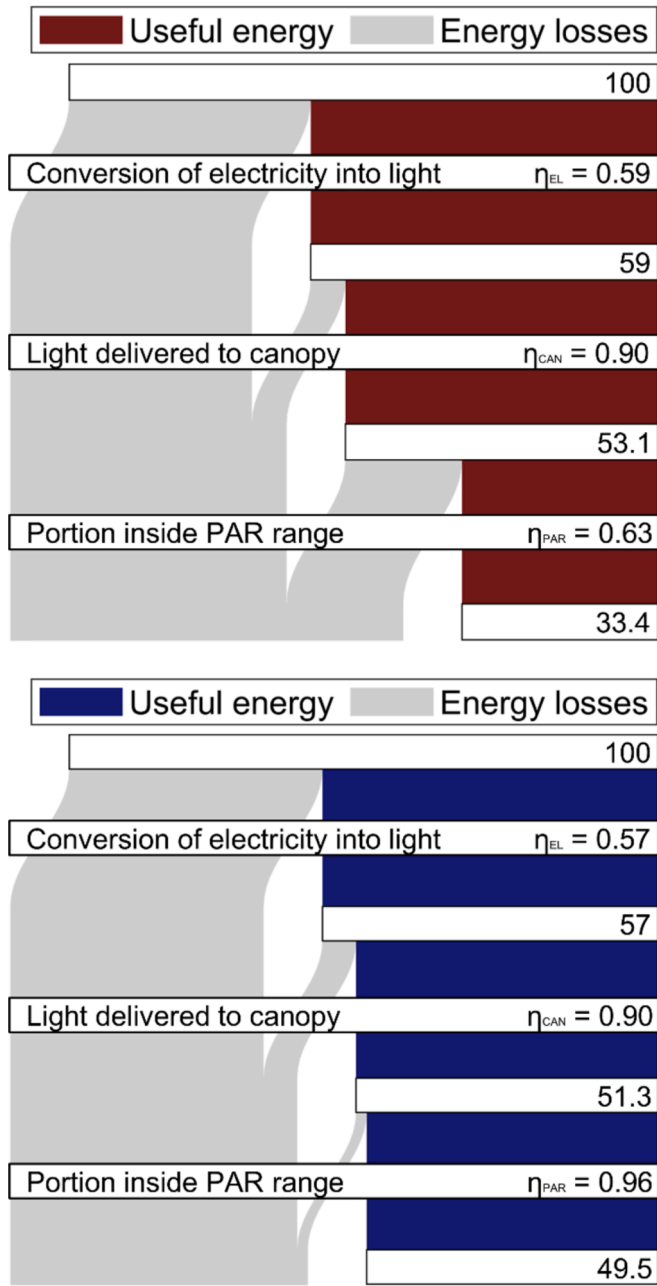


Fig. 6. Conversion of electricity into PAR: HPS (top) and LED (bottom). Data from [47,48].

4. Energy use for climate control

For climate control, the energy balance of the conditioned space is influenced by several aspects. The instantaneous energy balance of a greenhouse, expressed in W, can be described by Equation (3), where Q_{CC} is the energy power consumption for climate control, η_{HVAC} is the conversion efficiency of the Heating, Ventilation and Air Conditioning (HVAC) system, $\frac{dE_{air}}{dt}$ is the energy of the air volume associated with a change in air temperature, Q_{light} is the energy generated by light sources, whether solar or artificial, Q_{env} is the energy exchanged with the external environment through the envelope, Q_{vent} is the energy exchanged through ventilation, Q_{plant} is the energy associated with plant activity, and Q_{FIR} is the energy associated with long wave or far infrared radiation. For the interpretation of each term, reference is made to [54].

$$Q_{CC} = \frac{\frac{dE_{air}}{dt} - Q_{light} + Q_{env} + Q_{vent} + Q_{plant} + Q_{FIR}}{\eta_{HVAC}} \quad (3)$$

Vertical farming introduces two radical changes in the energy balance of the indoor environment compared to greenhouses. On the one hand, the opaque, thermally insulated and airtight walls make the terms Q_{env} , Q_{vent} , and Q_{FIR} negligible; on the other hand, the relevance of the terms Q_{air} and Q_{light} is modified since the air volume is significantly reduced and the variable sunlight is substituted with constant artificial lighting during light hours. This is summarised in Eq. (4), where ρ and c_v are the density and specific heat of air, V is the air volume, dT/dt is the temperature variation per unit time, E_{PAR} is the power lighting requirement in terms, h_{ev} is the latent heat of evaporation of water, g_e is the evapotranspiration flowrate, and X_c and X are the water concentration in crop and air.

$$Q_{CC} = \frac{\rho \cdot V \cdot c_v \cdot \frac{dT}{dt} - \frac{(1-\eta_L)}{\eta_L} \cdot E_{PAR} + h_{ev} \cdot g_e \cdot (X_c - X)}{\eta_{HVAC}} \quad (4)$$

Assuming a constant temperature, and thus neglecting the transition periods highlighted in, energy use depends only on the efficiency of the HVAC system, on the electricity used for lighting, and on the plant physiological activity. This means that, unlike greenhouses, the energy demand of the conditioned space in vertical farming is independent of the outdoor environment. Except for transition periods, it is claimed that vertical farms can be modelled as two states, the light and dark period [55]. The ratio between the thermal generation of the light sources and the evapotranspiration load of the plants becomes a relevant design parameter [56]. For example, in [57] it is noted that a halving of the evapotranspiration rate significantly modifies the energy requirements of a vertical farm, almost eliminating the heating demand. However, considering that ratio as a constant is a relevant simplification, as evapotranspiration increases with LAI [58].

Fig. 7 shows the variation in normalised average daily loads in the conditioned environment during an optimised growth cycle from Ref. [58], and adapted for the optimal lighting schedule calculated in Ref. [53]. The thermal generation due to lighting increases with crop growth to match the increased photosynthetic capacity, allowing to reduce the cooling requirements compared to a fixed lighting schedule. Dehumidification increases due to increased plant activity, contributing to reducing the cooling demand during light periods and to increasing the heating demand during dark periods.

In addition to energy loads, the performance of the HVAC system is important in characterising equipment consumption. Compared to lighting, the number of functions and devices involved in climate control is much higher, resulting in a larger operational variability that makes the evaluation of an overall efficiency more complex.

Considering the heat pump technology, COP is highly dependent on machine design and operation, causing nominal performance to deviate from the ideal performance, as in the case of dehumidification [59]. As shown in Fig. 8, heat pump operation is influenced by both external conditions and operating parameters, with important consequences on the deviation from nominal performance in greenhouse heating [60], greenhouse cooling [61] and vertical farm cooling [47]. Accordingly, while climate control loads become independent from the external temperature in vertical farms, there is still a strong influence on the HVAC energy consumption.

5. Energy benchmarks for vertical farming

The analysis carried out has highlighted the complexity of energy use in vertical farming. It has shown that to fully understand energy use, it is essential to consider crop requirements, related to light demand and latent loads, vertical farm configuration, related to visual and inertial thermal effects, and energy system design and operation, related to the

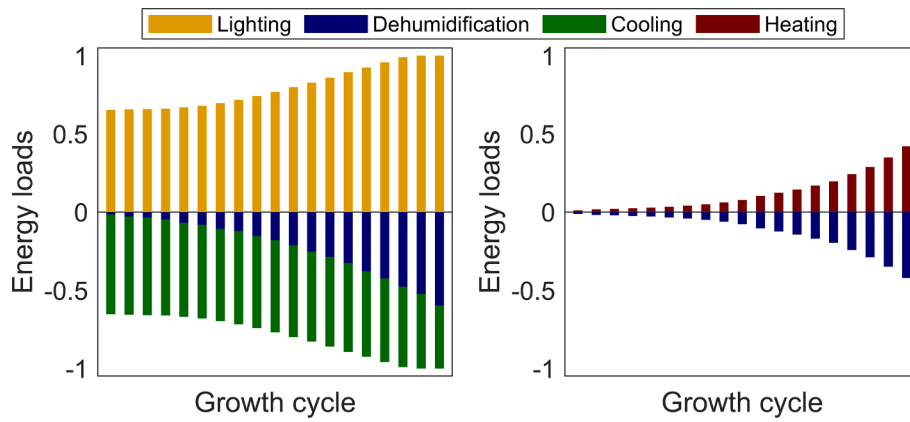


Fig. 7. Normalised climate control loads during light (left) and dark periods (right). Data from [53,58] for temperature of 24 °C, photoperiod of 16 h, PPFD of 400 $\mu\text{mol}/\text{m}^2 \text{s}^{-1}$, and relative humidity of 80 % and 89 % during light and dark periods.

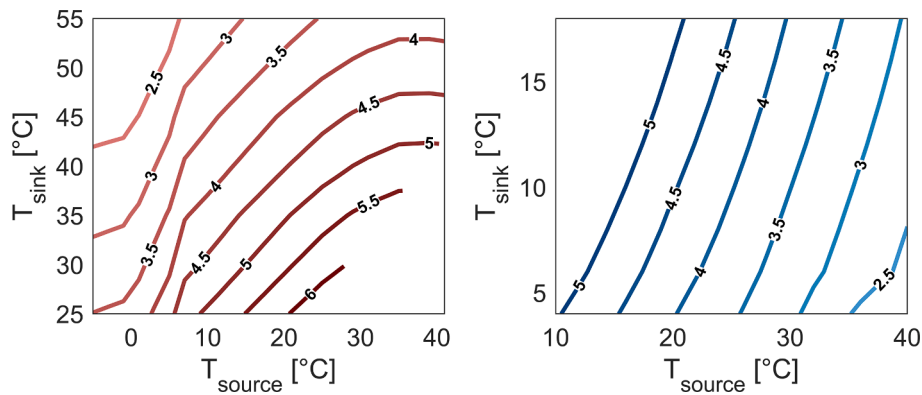


Fig. 8. Heat pump COP: heating (left) and cooling (right).

conversion of electricity or other energy carriers into light and thermal energy.

5.1. Current benchmark

The literature on energy information in vertical farming is sparse. Energy is sometimes reported in absolute terms [57] but more often properly related to production and space occupation. The Specific Energy Consumption (SEC) relates energy to a measure of production and is expressed in kWh kg^{-1} , as typically done for industry [19], whereas the Energy Use Intensity (EUI) relates energy to a spatial measure and is expressed in $\text{kWh m}^{-2}_{\text{footprint}} \text{year}^{-1}$, as typically done for buildings [62]. Several variations of these indicators can be used, including $\text{kWh kg}^{-1}_{\text{dry matter}}$ [55], $\text{kWh m}^{-2}_{\text{canopy}} \text{year}^{-1}$ [63], etc.

For standardisation purposes, it is important that information on production, environment, and equipment is accurately reported along with energy use. Production information defines the object of analysis, including information that is useful for harmonising performance indicators. Environmental information is the primary driver of energy use and includes measures of light, temperature, relative humidity, and CO_2 concentration. Equipment information includes both passive and active elements, such as the module layout, or the components and control strategies of the lighting and HVAC system.

For the identification of the benchmark, some literature studies where the focus is on energy use have been included in Table 1. Only studies dealing with both lighting and climate control and reporting relevant contextual information on all the aspects previously mentioned have been considered. The quality of reporting varies between studies, with some aspects, such as equipment, being completely neglected in some cases.

The total SEC reported in Table 1 varies considerably from around 5 kWh kg^{-1} to slightly less than 40 kWh kg^{-1} . Due to its suitability for vertical farming, lettuce is the crop analysed in 14 out of the 16 studies considered. As illustrated in Fig. 9, the annual yield embodies the relationship between the SEC and the EUI, with values ranging from around 50 $\text{kg m}^{-2} \text{year}^{-1}$ to 120 $\text{kg m}^{-2} \text{year}^{-1}$. With the exception of an experimental study [64], most values are concentrated in the range of 10–18 kWh kg^{-1} . This is consistent with the range of 7–15 kWh kg^{-1} reported in [65], of 10–20 kWh kg^{-1} found in [25] and with the performance of real plant factories, as a large-scale commercial vertical farm achieved an energy performance of 9.9 kWh kg^{-1} [66].

More in-depth analysis should focus on the investigation of scale effects in energy use, distinguishing between large vertical farms and small container farms and the associated envelope and equipment solutions. Without distinguishing between greenhouses and vertical farms, a report on controlled environment agriculture has evaluated that small

Table 1
Literature data on vertical farming.

Ref.	Type ^a	Energy					Production			Environment					Equipment			
		Total	Lighting	HVAC	Lighting	Crop	Yield	Growing	Plant	PPFD	DLI	Photoperiod	Temperature	Relative	LED	LED	Boiler	Heat pump
		SEC [kWh kg ^{−1}]	SEC [kWh kg ^{−1}]	SEC [kWh kg ^{−1}]	share [%]	[-]	[kg m ^{−2} canopy year ^{−1}]	area [m ²]	spacing [head m ^{−2}]	[μmol m ^{−2} s ^{−1}]	[mol m ^{−2} day ^{−1}]	[h]	[°C]	[%]	efficacy [μmol J ^{−1}]	efficiency [%]	[-]	COP [-]
[57]	S	19.5–22.9 ^b	9.8	9.8–13.1 ^c	43 – 50	Lettuce		1712	48	197	17	24	24/19		1.7		0.8	Constant, 5.5
[17]	S	3.3 ^d	3.1 ^d	0.1 ^d	97	Tomato	74		4	427	20	13	17–28	60–90	^d	^d		3.1 heating,
[70]	S	19.2	19	0.2 ^e	99	Lettuce	53.4	1171			13		18.3–21.1		1.3	28	0.8	3.93 cooling ^d
[64]	M	38.5 ^f	30.3 ^f	8.2	79	Lettuce	69.5	338							^f	^f		
[55]	S	17.0–19.0	15.8	1.2–3.2	83–93	Lettuce	71.4	50,000		500	28.8	16	24–30	65/90	2.7	52		3.4 – 10
[68]	S	7.3–8.4 ^g	6.6–6.9 ^g	0.7–1.5 ^g	82–90 ^g	Lettuce	79.6–87.9	1360	48	226, 260	13, 15	16	24/19		2.0, 2.5, 3.0	63	0.8	Constant, 3.45
[38]	S	29.7 ^h	26.4 ^h	3.3 ^h	89	Cai Xin and Xian Cai	88.8, 135.2	242	47.6	619, 857	29.1	11	28	80				
[20]	S	15.6–20.1	14.2	1.4–6	71–91	Vegetable	38	3000		250	13.7	14	22.1		2.7	59		Temperature-dependent
[20]	S	9.8–12.6 ⁱ	8.9 ⁱ	0.9–5.5 ⁱ	71–91	basket Lettuce	52.6	3000		200	11.5	16	18		2.7	59		Temperature-dependent
[21]	M	14.7	9.7	5.0	66	Lettuce	69.1	66	43	140.3	10.1	20					0.9	4
[21]	M	30.7	21	9.7	66	Basil	36.8	56		150	10.8	20					0.9	4
[71]	S	18	9.4	8.6	52	Lettuce	61	64.2	18	434	14	9	21/18	70/74	2.6	37	1	3.66
[72]	S	6.2–12	4.6 – 8.3	1.1–3.7	58–84	Lettuce	103.6	20,475	48	162–197	14, 17	24	19–24	50–85	2.0, 2.5, 3.0			Temperature-dependent, 4
[69]	S	5.3–14.1 ^b	3.5–9	2.2–5.1	64–80	Lettuce		18	33.3	250	10.8	12	24/20	75–85	1.17, 1.5, 2, 2.5, 3			Part-load-dependent, 3
[24]	S	13.7–14 ^j	9.5–10 ^j	4.0–4.4 ^j	68–83 ^j	Lettuce	75 ^j			{250:75: 700}	21.6	16	{22:1:28}	75	2.3	{35:5:60}	0.9	Temperature-dependent
[23]	S	4.4, 7.3, 9.5	3.5, 5.9, 7.6	1.1, 1.4, 1.9 ^k	80	Lettuce	91, 114, 127		25	200, 400, 750	11.5, 23, 42	16	24		3.5	65	^k	^k
[22]	S	10.2–14.7 ^j	4.2–7.0 ⁱ	6.0–7.7 ⁱ	41–48 ⁱ	Lettuce	87–164 ⁱ			{200:100:700}	11.5–40.3	{12:2:22}	{20:4:28}	65–85/79–87	2.6	52		
[26]	S	9.5–10.5 ^m	6.4–7.1 ^m	3.1–3.6 ^m	64–70 ^m	Lettuce	97.8	80		270	15.6 ^l	16	22–25/18–20 ^l	70–80 ^l	2.7	52		3–6

^a S, simulated; M, measured.

^b Harmonisation is performed considering a yield equal to [72].

^c Large heating share due to poorly insulation.

^d HPS lamps are considered, with an unrealistic efficacy of 9.1 μmol J⁻¹, as noted by [20]. Thermal energy generation appears to have been neglected.

^e No active cooling is considered, only mechanical ventilation.

^f Fluorescent lamps are used.

^g Reference values for 2.5 μmol J⁻¹ of LED efficacy.

^h For this study, [21] assumed a lighting energy use of 26.4 kWh kg⁻¹ and [20] assumed a total energy use of 59.1 kWh kg⁻¹. Here, the former was considered with the original lighting share.

ⁱ Energy consumption was evaluated by averaging between the reported savings during winter and summer for total and HVAC energy use. The calculated lighting use is consistent with a proportion based on yield and DLI.

^j Energy uses refer to the combination of PPFD 250 μmol m⁻² s⁻¹, temperature 24 °C, and LED efficiency 60 %. Lighting share also includes the variation of PPFD.

^k HVAC energy use is estimated by considering an additional 25% to lighting.

^l Energy uses refer to the temperature-averaged values and to PPFD 300 μmol m⁻² s⁻¹.

^m Data refer to the reference condition of fixed setpoints.

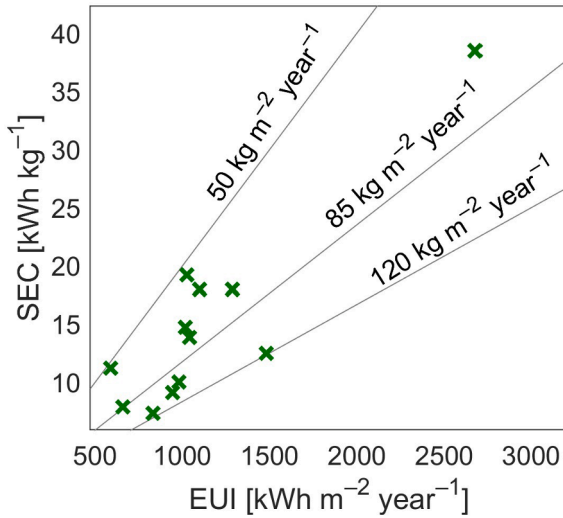


Fig. 9. SEC and EUI for lettuce production in vertical farming.

(<1000 m² of growing area), medium (1000–5000 m²), and large (>5000 m²) facilities have an average energy consumption of 34.2 kWh kg⁻¹, 14.9 kWh kg⁻¹, and 8.3 kWh kg⁻¹, respectively [67]. While this can be mostly attributed to scale effects in greenhouses due to a decreasing ratio of surface to volume, no significant scale effect is observed from the studies considered. This can be attributed to the limited impact of surface-related terms on the energy balances of vertical farms, such as the significant reduction of thermal dispersion.

Lighting energy consumption is strongly dependent on LED efficacy and efficiency, with large differences in the values considered, ranging from 1.3 μmol J⁻¹ to 3.0 μmol J⁻¹ and from 28 % to 63 % respectively. Advanced LEDs can reduce energy consumption by more than 50 %. Both constant and variable COP were considered for climate control, with differences depending on the location. The presence of an air economiser was one of the most important HVAC features, resulting in a shift from heating to cooling [68] and a reduction in HVAC energy consumption of more than 50 % [69]. Given this variability, the resulting lighting share ranges from slightly less than 50 % to almost 100 %, with typical values around 65–90 %.

5.2. Technical benchmark

The identification of an energy benchmark that incorporates technological progress in equipment efficiency and operational control strategies is essential to determine the potential development of such innovative agricultural technique, as it facilitates comparison with conventional agricultural techniques and provides insight into the current energy performance gap.

By combining Eq. (1) and Eq. (3), a simple formulation for the lighting benchmark is obtained as a function of crop and equipment features. By considering an harvest index of 85 % and dry matter content of 3.5 % for lettuce [23] and assuming that maximum conversion efficiencies are achieved in vertical farming, the resulting lighting energy requirement is 2.3 kWh kg⁻¹.

Table 2 shows the lighting benchmark for lettuce production under varying LED efficiencies. The values range from 4.7 kWh kg⁻¹ to 2.6 kWh kg⁻¹ for LED efficiencies of 0.5 and 0.9. These results are consistent

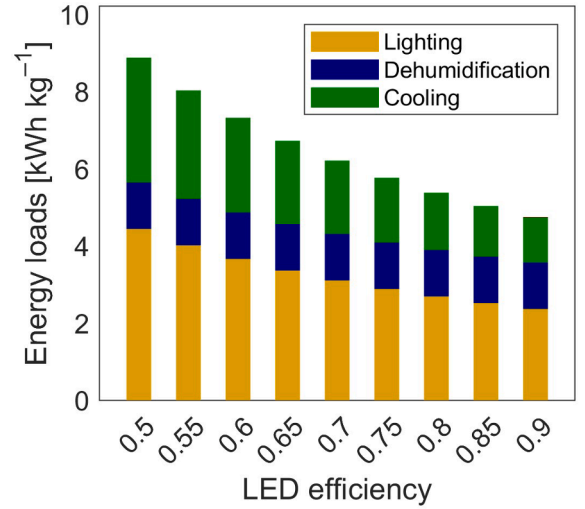


Fig. 10. Climate control loads during light periods under varying LED efficiency.

with those reported in Ref. [23], which are based on the experimental results of Ref. [35].

$$E_L = \frac{E_b}{\eta_{int}} \cdot \frac{DM}{HI} \cdot \frac{1}{\eta_L} \quad (6)$$

The proposed approach based on the assumptions on the energy efficiency of crop internal processes yields similar results to the more common approach on a photon basis [73], confirming the validity of the rule of thumb for the energy content. By adapting the quantitative values in Eq. (6), the calculation of the energy benchmark may be performed also for other crops.

The associated climate control energy requirement can be obtained by adjusting the climate control loads in Fig. 7 for varying LED efficiency. This requires modification of the thermal generation due to lighting under fixed dehumidification demand, since the environmental conditions are assumed not be changed. The resulting energy loads during light periods are shown in Fig. 10, while those during dark periods remain unchanged. Considering an overall COP of the HVAC system between 2 and 6, a benchmark between 0.5 kWh kg⁻¹ and 2.5 kWh kg⁻¹ is obtained for climate control.

Fig. 11 shows the resulting energy benchmark and lighting share for lettuce production under varying LED efficiency between 0.5 and 0.9 and varying heat pump COP between 2 and 6. The resulting energy benchmark varies between 3.2 kWh kg⁻¹ and 7.4 kWh kg⁻¹, with LED efficiency having a much greater influence. The lighting share varies between 64 % and 85 %, with a greater influence of the HVAC COP.

The obtained energy benchmark for current LED efficiency is much lower than most values reported in Table 1, highlighting a relevant energy performance gap that suggests a large potential for improvement. For lighting, the main source of deviation is the application of constant lighting schedules rather than optimal ones. In quantitative terms, this introduces 25 % additional energy consumption due to excess of light in the earlier stages of plant growth, which corresponds to an increase of 24 % in the total energy consumption due the associated cooling increase. Anticipated improvements in LED efficiency from 0.5 to 0.9 are expected to decrease the lighting energy consumption by up to 45 %,

Table 2
Lighting SEC for lettuce under varying LED efficiency.

LED efficiency	0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.87	0.90
This study	4.7	4.2	3.9	3.6	3.3	3.1	2.9	2.8	2.7	2.6
[23]				3.7					2.7	

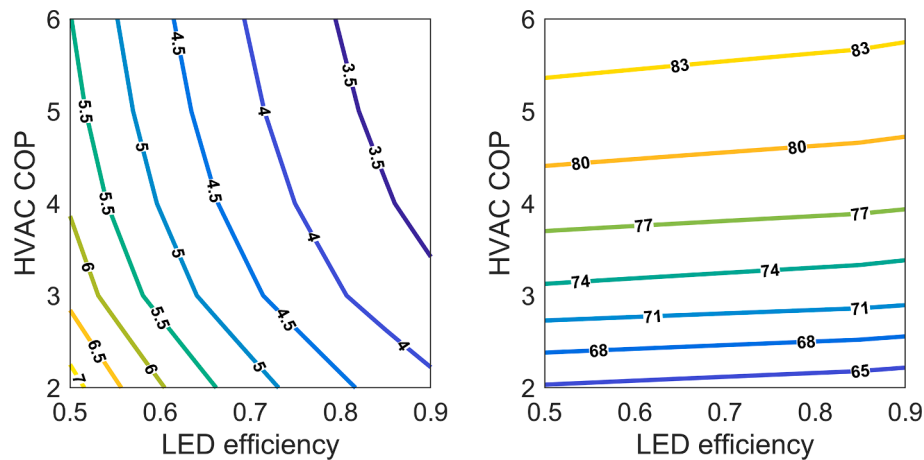


Fig. 11. SEC (left) and lighting share (right) for lettuce under varying equipment efficiency.

which corresponds to between 28 % and 38 % of the total energy consumption. Improvements in the COP of the HVAC system from 2 to 6 can lead to reductions in climate control energy consumption of 66 %, which corresponds to between 28 % and 30 % of the total energy consumption. This can be mainly achieved through a supervisory control of the components and the implementation of an air economiser.

6. Concluding discussion

This contribution establishes a foundation for standardising energy performance metrics, providing actionable targets for operators and policymakers in vertical farming while enabling more consistent comparisons with traditional agricultural systems.

6.1. Standardisation of energy performance

Modelling studies and existing facilities have been used to define a current energy benchmark, whereas expected advances in equipment efficiency and optimal control strategies have been incorporated in a technical energy benchmark.

The technical benchmark for energy performance in vertical farming is developed through a straightforward approach that avoids the complexity of detailed models. Lighting requirements are based on assumptions about the energy content of biomass and PAR conversion efficiency. Results align with more typical evaluations based on photons

and carbon conversions. This should be validated, especially when applying the methodology to other crops. The climate control requirement is derived using the energy balances from a literature study [58], which refer to specific environmental conditions. For purposes of generalisation, other combinations of environmental conditions and other crops may be tested.

The combination of the performed literature review and the identified benchmark offers a valuable contribution to the standardisation of energy performance, which is summarised in Fig. 12, distinguishing between modelling studies, existing facilities, and expected technical limits for lettuce production in vertical farming.

Data from existing facilities align well with modelling studies, with the exception of an experimental vertical farm used for scientific purposes where energy efficiency is not the main goal [64]. A representative range of the SEC equal to 10–18 kWh kg⁻¹ is identified, which corresponds to values of EUJ values between 850 and 1150 kWh m⁻² year⁻¹. The technical benchmark resulting from optimal lighting schedules, improved LED efficiency and climate control COP is equal to 3.1 kWh kg⁻¹ and 7.4 kWh kg⁻¹. This is on average three times lower than modelling studies depicting realistic conditions, indicating a large room for improvement. While the benchmark of modelling studies and existing facilities reflects a realistic current performance standard that can be achieved by implementing best practice, the technical benchmark sets an upper limit that can be achieved only upon technological innovation.

6.2. Implications for sustainability

The definition of energy benchmarks for vertical farms allows comparison with traditional agricultural techniques, although defining an energy benchmark for greenhouses is challenging due to the wide variability in climate conditions and the associated technological packages. The SEC for lettuce production in greenhouses ranges around 5.3–33.3 kWh kg⁻¹ [16], though lower values can be achieved in unheated tunnels with no energy consumption or in highly efficient heated greenhouses (around 3 kWh kg⁻¹) [23]. This comparison suggests that vertical farming may become competitive in terms of energy efficiency, though it does not account for productivity improvements, which could be better assessed using the EUJ.

To fully assess the overall energy sustainability of vertical farming, it is crucial to consider factors beyond energy efficiency. These include the shortening of supply chains [23], the economic costs and the environmental impacts of electricity use. While vertical farms may be more efficient than conventional agricultural techniques, they require more purchased energy [55].

The integration of renewable energy sources raises intriguing questions regarding future developments of photosynthetic and photovoltaic

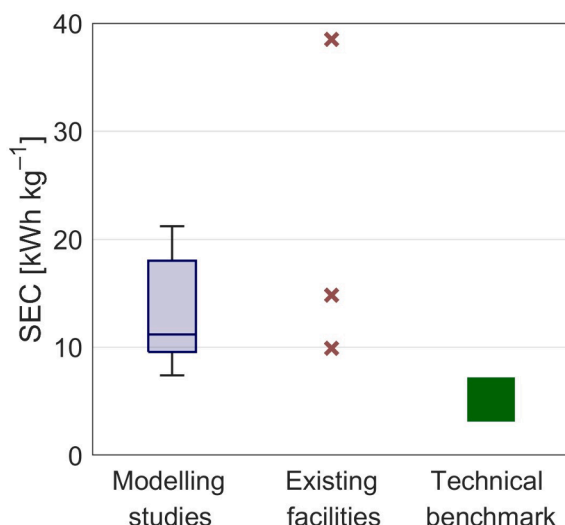


Fig. 12. Performance standards for lettuce production in vertical farming.

efficiencies [74]. The significant reductions of land use make the area available for photovoltaic modules insufficient to cover the increased energy demand. This suggests extending the boundaries beyond the boundaries of vertical farms to the upstream energy system, making the energy sustainability of controlled environment agriculture highly dependent on variations in electricity prices and carbon footprint. While vertical farming may compete with conventional agricultural techniques in terms of energy efficiency, the potential uptake of vertical farming will depend on carefully balancing the delicate trade-offs between energy use and other resources, considering economic cost and environmental impact, as well as local availability.

CRedit authorship contribution statement

Lorenzo Miserocchi: Writing – review, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Alessandro Franco:** Writing – review, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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